Housing First or Treatment First?

A stylised model, before the quantitative one

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Abstract

The smallest model in which the paper's two questions can be asked. Who uses drugs, and why are users poor? And what does housing assistance do to drug use? Three periods, one asset, and households that differ in permanent income and in nothing else. It is deliberately qualitative: it establishes directions and the sign and relative size of two opposing forces. Every magnitude in the paper comes from the full model that follows.

1 What this model is for

The full model has six state variables and roughly forty parameters. That is what the welfare and fiscal questions require, and it is far too much machinery for a reader to see which piece does the work. This note strips the environment to the smallest one in which two questions are still well posed.

1. **Who uses.** Drug use is concentrated among low income households. What produces that, and how strong does it have to be?
2. **What housing assistance does.** A voucher pulls drug use in two directions at once. Which force is larger?

Nothing else. No welfare, no government budget, no general equilibrium, no validation section. The calibration is rough on purpose. Section 6 justifies every component by removing it, and section 7 says what the model cannot do and why none of it is an accident.

2 The environment

Three periods. A household has permanent income y, fixed, lognormal, and differs from other households in nothing else.

Period 1. Employed, in a rented unit. Choose consumption, drug use $v \geq 0$ and saving, with flow utility

$$\log c \;+\; \gamma \log h \;+\; \theta\big[\log(b+v) - \log b\big].$$

Using risks a substance use disorder next period, with probability $1 - e^{-\lambda v}$.

Period 2. The disorder d is realised. The household is unemployed with probability u_0, or the higher u_1 if $d = 1$. Income support replaces half of previous earnings, capped at half of average income. Choose housing and saving. Recovery is ρ.

Period 3. Terminal, same risks.

Housing. A large unit, a small unit, or the street. The street is free but supplies only h_{hl} of housing services. A voucher arrives with probability π_V and cuts the tenant's rent.

The two policies. Under Housing First the voucher arrives whatever the household's disorder status. Under Treatment First only if it is clean. That single difference is the entire policy contrast, and it is what makes the decomposition in section 5 possible.

No preference heterogeneity. Households differ in permanent income and in nothing else, so every household at a given income makes the same choice and the drug use decision is a *threshold in income* rather than a probability. The sharpness is the point. It also keeps intact the paper's claim to generate the income gradient without ex ante preference heterogeneity: there is none here to do the work.

3 What is calibrated

Three parameters, bracketed on three targets.

parameter	target	data	model
b, the pull of the first unit	drug participation rate	5.13%	4.51%
h_{hl}, housing on the street	homelessness rate	0.399%	0.20%
λ, the addiction hazard	users who acquire the disorder	33.2%	33.2%

Everything else is set. Rent is set so that the standard unit costs 21.2 per cent of average income, which is the housing budget share in the data. Income support replaces half of previous earnings capped at half the average, which is roughly what unemployment insurance does. The unemployment rate given the disorder is the measured 16.25 per cent from NSDUH. The housing weight is 0.25, and the discount factor, interest rate, survival, recovery and the mortality multiplier take standard values.

Bracketing rather than a joint fit, for a reason worth stating because it recurs in the full model. With no preference heterogeneity the aggregate is a cliff in b: participation is one below a narrow band and zero above it, so a gradient method steps out of the band, finds a flat surface and walks to a corner. Each parameter is therefore bracketed on its own monotone target in sequence.

4 Who uses

Start from the force that has to be overcome. Drugs are a normal good bought at a price, so a household with twice the income gives up less utility per dollar spent on them. With *no consequences at all* for acquiring a disorder, drug use therefore concentrates at the top: the average user earns 3.4 times the population average. The model has to be forced to put drug use at the bottom; it does not go there on its own.

Two things could do the forcing.

mortality	the disorder kills you
unemployment	it costs you a job, income support cannot cover the rent, so you may end up on the street, where housing services collapse

Each is scaled from nothing to its calibrated value, and at every point b is rebracketed so participation stays at 5.13 per cent. The amount of drug use is therefore identical all along both curves, and the only thing moving is *which* households use.

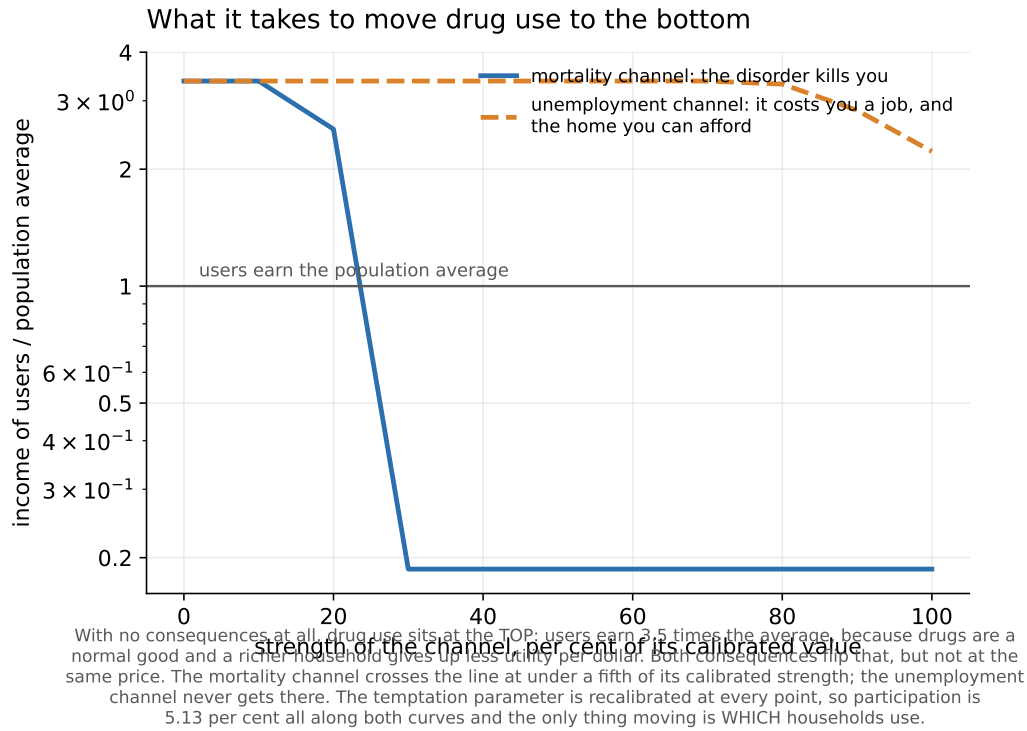


Figure 1: Only the mortality channel moves drug use to the bottom of the income distribution.

	mortality channel		unemployment channel	
strength	b	users / average	b	users / average
0%	0.2368	3.37	0.2368	3.37
25%	0.0090	0.19	0.0655	3.37
50%	0.0054	0.19	0.0337	3.37
75%	0.0038	0.19	0.0193	3.37
100%	0.0030	0.19	0.0128	2.22

The top row is the case with every consequence removed. Both columns start there, and users earn 3.4 times the average.

The mortality channel flips the gradient at a quarter of its strength. The unemployment channel never flips it at all. Even fully switched on it leaves users at 2.2 times average income, still above the population mean. That is a sharper statement than the paper has made before: it is not that mortality matters more, it is that in this environment the unemployment channel *cannot* produce the fact on its own however strong it is made.

The b column shows the recalibration doing real work. With no consequences a household needs almost no temptation and $b = 0.237$ delivers 5 per cent participation. Once the mortality channel is on, the same 5 per cent needs b near 0.003, eighty times more temptation. Without rebracketing b at every point the curves would be comparing economies with different amounts of drug use, and the exercise would say nothing about who uses.

Why the mortality channel wins, and it is not obvious. A penalty proportional to income scales with the thing it is meant to offset, so it cannot outrun the fact that drugs are cheaper in utility terms for the rich. A mortality penalty can, because what it destroys is a continuation value, and the continuation value rises with permanent income. The household with more to lose loses more.

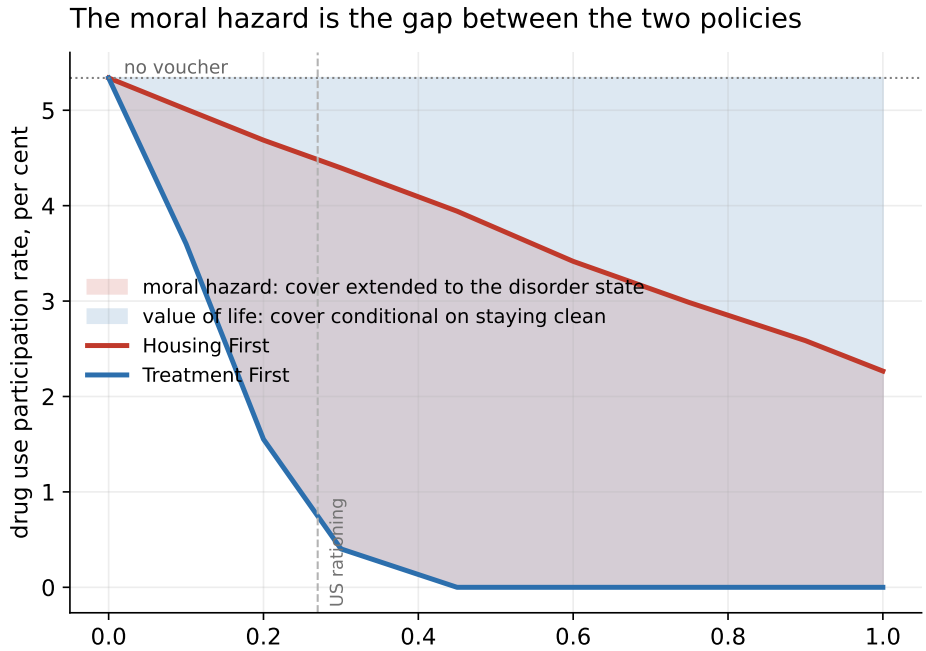
5 What housing assistance does

A voucher does two things to a household deciding whether to use.

It raises the value of staying clean, because a housed future is worth more than a homeless one and using puts that future at risk. Call this the value of life effect. It pushes use *down*.

It also cushions the consequence of acquiring the disorder, because the household that acquires it is housed anyway. Call this the moral hazard. It pushes use *up*.

The two policies separate them exactly. Under Treatment First the voucher is withdrawn on acquiring the disorder, so only the first force operates. Under Housing First it arrives regardless, so both do. The only thing Housing First adds to Treatment First is cover in the disorder state, so the gap between the two policies *is* the moral hazard.



Under Housing First a voucher is offered to anyone who is homeless, whether or not they are using drugs. Under Treatment First the household is clean. The only thing the first adds to the second is cover in the disorder state, so the red band IS the moral hazard. Both forces grow with generosity and they work in opposite directions, which is why the net effect of housing assistance on drug use is not signed by theory.

Figure 2: The moral hazard is the gap between the two policies.

Reading the numbers. All three are percentage points of the population drug participation rate, measured against the same starting point: no voucher programme at all, where 5.34 per cent of households use.

	participation	
no voucher at all	5.34%	
universal Treatment First	0.00%	value of life effect: −5.34 pp
universal Housing First	2.27%	moral hazard: +2.27 pp
net effect of Housing First against no programme		−3.07 pp

Move from no programme to universal Treatment First and participation falls by 5.34 points, to nothing: guaranteeing housing to the clean makes staying clean worth much more. Move instead to universal Housing First and participation is 2.27 points *higher* than under Treatment First. That difference is the moral hazard, and it is exactly the price of also covering households that already have the disorder, because that cover is the only difference between the two policies. The net effect of Housing First against no programme is −3.07 points: it still lowers drug use, because the first force is larger than the second.

Both forces are first order, and they work against each other. That is the central result of this section and the reason the paper needs a quantitative model. Theory does not sign the effect of housing assistance on drug use. Anyone asserting that housing the homeless must

increase drug use, and anyone asserting it must reduce it, is making a quantitative claim while sounding as though they are making a theoretical one.

Unconditional assistance still lowers drug use. Worth stating plainly, because it cuts against the intuition that unconditional help must raise use through moral hazard. The moral hazard is real and it is large. It simply loses to the other force.

Conditionality keeps the winning force and removes the losing one. That is the argument for Treatment First in one sentence, and it explains something about the previous version of this paper. Its weakest experiment imposed conditionality while holding the number of vouchers at the current rationed level. Both forces scale with generosity, so at the left of the figure there is very little for conditionality to work with while the cost of excluding people from a scarce benefit is immediate.

6 Why each component is here

Every component was tested by removing it, *fully recalibrating* the model without it, and re-running both counterfactuals. Recalibrating is the whole point: a component can look essential merely because removing it throws the model off its targets, and it can look irrelevant because the other parameters were never given the chance to compensate. Only refitting distinguishes the two.

removed	users / avg	value of life	moral hazard
nothing	0.17	−4.3	+1.5
the disorder kills you	3.37	−1.3	+2.5
the disorder costs you a job	0.19	−6.6	+2.3
the street raises mortality	0.17	−4.5	+1.6
recovery is harder on the street	0.17	−4.3	+1.5

The disorder kills you. Kept. Removing it sends users from 0.17 to 3.37 times average income, straight back to the top of the distribution. This is the single most important component in the model and the one the paper's mechanism rests on.

The disorder costs you a job. Kept, for the second result rather than the first. It barely touches who uses, moving the ratio from 0.17 to 0.19. But removing it changes the value of life effect from −4.3 to −6.6 points, because unemployment is what makes the disorder threaten your housing, and housing is what the voucher insures. Without it the policy has much less to do.

Income support that cannot cover the rent. Structural. Not a component to be justified separately, because it is what unemployment pays. Its level is the most consequential number in the model. Set it above the rent on the small unit and every household can always afford a home, the disorder never threatens the roof over anyone's head, the voucher has nothing to

insure, and the moral hazard is exactly zero. A replacement rate of half of previous earnings capped at half the average leaves the poorest unable to house themselves out of income alone, which is what connects the disorder to homelessness.

Two rental sizes. Kept. With a single size a household whose cash falls below the rent cannot be housed at all, whatever the street is like, and the model puts the entire bottom of the distribution on the street. With a small unit available they downsize instead, which is what happens in the data and what the full model's housing grid is for.

The street is a much worse home. Structural. Without it there is no homelessness to explain and no voucher worth having. It is also the parameter bracketed on the homelessness rate, so it cannot be ablated: removing it and then recalibrating simply restores it.

The street raises mortality. Removed. Neither result moves. Homelessness is a fifth of one per cent of the population here, so a channel acting only on the homeless cannot move an aggregate.

Recovery is harder on the street. Removed. Same reason, same numbers.

Both removed channels stay in the *full* model, where mortality and welfare are the objects of interest and the counterfactuals move the homeless population a long way. They are absent here because this model answers two questions and they bear on neither.

7 What this model cannot do

None of this is accidental, and none of it is fixable without adding the machinery the model exists to avoid.

The price elasticity of drug participation. With every household at a given income identical, a change in the price of drugs moves the whole threshold and takes a large mass with it. The elasticity is pinned by the density of the income distribution at the threshold rather than by any parameter, and it comes out between three and four against roughly 0.45 in the data. Matching it needs dispersion in tastes, which is the thing deliberately excluded.

Drug spending as a share of income. With one taste, optimal use solves $\theta/(b+v)$ equal to marginal cost exactly, so the level follows once b is set.

The magnitude of the income gradient. Users earn 18 per cent of the average here against 95 per cent in the data. The direction is right and the comparison across channels is informative; the level is not.

Anything requiring aggregation. No government budget, no market clearing, no welfare. A drug supply curve was built into an earlier version and removed after switching market clearing off changed neither result to two decimal places. It belongs in the full model, where the fiscal and general equilibrium consequences it interacts with actually exist.

8 What the full model adds

Six state variables instead of one, so heterogeneity comes from the environment rather than from assumed taste differences. A government budget constraint, so policies have to be paid for. An equilibrium drug price. Serious mental illness. An endogenous treatment decision, so that a policy named Treatment First contains treatment. And welfare, which is what the policy question is actually about.

Two things here carry forward as questions rather than answers.

The first is whether the unemployment channel remains unable to produce the income gradient once it is given the richer environment: more states, a budget constraint, and counterfactuals that move homelessness a long way. The full model's channel decomposition has to answer that rather than assume it.

The second is whether the ranking in section 5 survives. Here the value of life effect beats the moral hazard by three points out of five. That both forces are first order is the robust finding. Which one wins is not.